Christian M. Lange

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Research group website Google Scholar PhD Defense

Profile-Summary

- Experimental physicist specializing in quantum optics and nanophotonics, with a strong track record of rapidly building end-to-end systems. Led a research subgroup from an empty optical table to a *Nature Physics* publication in 2.5 years. Deep expertise in nanofabrication, tabletop optics, programming and automation, quantum theory, and materials synthesis.

Education

2026 – Present
- **Postdoctoral Researcher**, Purdue University, Quantum Photonic Integrated Design Center (QuPIDc),
  Advisor: Dr. Libai Huang

2021 – 2025
- **Ph.D. in Physics**, Purdue University
  Thesis: *Quantum Light–Matter Interactions with Organic Molecules*
  Advisor: Dr. Jonathan D. Hood

2017 – 2021
- **B.S. in Physics**, Utah State University
  Minors: Mathematics, Computer Science
  Summa Cum Laude GPA: 3.99 / 4.00
  Thesis: *The reflection of light from periodic conducting interfaces* (2021)
  Advisor: Dr. Tsung-Cheng Shen

Publications

- **Lange, C. M.**, E. Daggett, V. Walther, L. Huang, and J. D. Hood, "Superradiant and subradiant states in lifetime-limited organic molecules through laser-induced tuning", *Nature Physics*, vol. 20, pp. 836–842, 2024. DOI: 10.1038/s41567-024-02404-4.
- E. Daggett, **Lange, C. M.**, et al., "Many-body entanglement in solid-state emitters", *Nature Reviews Materials*, 2026. DOI: 10.1038/s41578-026-00893-8.
- **Lange, C. M.** et al., "A hybrid molecular–nanophotonic platform for on-chip cavity quantum electrodynamics and collective interactions", *ACS Nano*, 2026. DOI: 10.1021/acsnano.5c19465.
- **Lange, C. M.** and T.-C. Shen, "Fabrication of height-modulated carbon nanotube forests: Morphologies and prospects for broadband absorption", *Carbon Trends*, vol. 4, p. 100 070, 2021. DOI: 10.1016/j.cartre.2021.100070.
- A. D. Keni, **Lange, C. M.**, et al., "Vapor phase assembly of molecular emitter crystals for photonic integrated circuits", Under review; submitted to *ACS Photonics*. DOI: 10.48550/arXiv.2602.18517.
- I. Agarwal, A. Kundu, **Lange, C. M.**, and J. D. Hood, "First order maxwell operator formalism for macroscopic quantum electrodynamics", Under review; submitted to *Phys. Rev. A*. DOI: 10.48550/arXiv.2603.27475.

Patents

- **Lange, C. M.** and J. D. Hood, *Photon emission with organic molecules and nanophotonics*, U.S. Provisional Patent Application, Purdue Research Foundation (PRF Ref: 2025-HOOD-71146-01), 2025.

Selected Projects

Nanofabrication

- **Si_3N_4 nanobeam cavities and grating couplers** — Designed/fabricated various cavity geometries with optical $Q \sim 10{,}000$ and mode volume $\sim (2.5\,\lambda/n)^3$ and grating couplers with 20% coupling efficiency.

Optics

- **Confocal microscopes** — Built several scanning confocal microscopes in different configurations (galvos vs. motorized stages and free-space vs. optical fiber spatial filtering).
- **Time-domain quantum emitter measurements** — Photon correlation Fourier spectroscopy, Hanbury Brown-Twiss measurement, and fluorescence lifetime of individual quantum emitters.
- **Lasers & cavities** — Aligned several MSquared Ti:sapphire lasers and high-finesse cavities.
- **Extreme cross-polarization extinction** — Developed precision optics for cross-polarization extinction with visibility > 70 dB.

Micromechanics

- **Crystal micromanipulator** — Positioned 30 μm crystals onto nanophotonic devices using a tapered fiber on a 3D micropositioner.
- **Motorized XY stage** — Built precision stage for heavy loads (> 25 lbs) with < 2 μm repeatability.

Programming, electronics, and automation

- **Experiment automation** — Python based GUI and custom drivers for remote/automated experiments: FAST Comtec and Swabian time taggers (Python), M Squared laser (TCP/IP), various optomechanical instruments (NI-DAQ, Python), and custom instruments (ARduino, RPi, ESP32).
- **Low-cost photon counter** — ESP32-based solution for low-time resolution photon counting.
- **Remote lab monitoring** — RPi/Arduino sensors + Grafana temperature monitoring.

Theory

- **Collective quantum systems** — QuTiP-based simulator for collective interactions.
- **Electromagnetic scattering** — Analytical and numerical modeling of reflection from periodic metal grids (undergraduate thesis topic).
- **Probability and linear algebra course development** — Assisted in developing courses for probability and statistics and linear algebra based on Jupyter notebooks.

Materials synthesis

- **DBT:anthracene crystal growth** — Built a custom two-stage tube furnace for vapor-phase growth of ultra-high aspect-ratio crystals and implemented a protocol for solution-based growth of DBT-doped anthracene nanocrystals.

Graphic design

- Various scientific graphics made in Blender, Inkscape, and Adobe Illustrator for publications and presentations.

Machining and 3D printing

- Basic 3D printing and use of end mills for making custom lab components.

Posters and Talks

Oral Presentations

Jan 2026 — *Cavity QED with individual cryogenic molecules*, SPIE Photonics West

Posters and Talks (continued)

Apr 2024	*Superradiant and subradiant states in lifetime-limited organic molecules*, Second Annual Birck Symposium.
	Workshop: A visual introduction to quantum optics, Quantum Science at Purdue.
Sept 2023	*Seminar: Superradiant and subradiant states in lifetime-limited organic molecules through laser-induced tuning*, Purdue AMO-QIS.
Oct 2022	*Organic dye molecules as a platform for coherent light–matter interactions*, Quantum Advances Seminar.
Apr 2021	*Electromagnetic scattering from periodic conducting arrays*, USU Student Research Symposium.
Dec 2020	*Modeling reflectance spectra of nanorod arrays by arrays of light sources*, USU Student Research Symposium.
Oct 2020	*Modeling reflectance spectra of nanorod arrays by arrays of light sources*, APS Four Corners Section.
Feb 2020	*How to create very dark surfaces for applications*, Utah Conference on Undergraduate Research.
Oct 2019	*Synthesis of carbon nanotube forests on deep-etched patterns for light absorption*, APS Four Corners Section.

Poster Presentations

Aug 2024	*Scalable collective systems through laser-induced frequency tuning of organic molecules*, Cooperative Phenomena in Hybrid Quantum Systems Workshop.
June 2023	*Coupling organic molecules to nanophotonic cavities to study collective effects*, DAMOP.
Apr 2023	*Tailoring the entanglement of quantum emitters through light-induced frequency tuning*, Purdue Quantum Summer School.
Oct 2021	*Scattering of light from periodic conducting structures*, APS Four Corners Section (co-author).
Apr 2019	*Fabrication of carbon-nanotube based bolometers*, USU Student Research Symposium.

Awards and Achievements

2024	1st Place Oral Presentation, Second Annual Birck Symposium.
2021–2025	Purdue University Ross Fellowship.
2021	Joyce Kinkead Outstanding Honors Capstone Award.
2020	PEAK Summer Research Fellowship.
	USU College of Science Scholarship.
	Lillywhite Scholarship.
	USU "A" Pin Award.
2019	Outstanding Student Presentation, APS Four Corners Section.
	URCO Grant (Utah State University).
	USU College of Science Mini-Grant.
	Sigma Pi Sigma (Physics Honor Society), Member.
	O. Harry Otteson Scholarship.
2018	Honorable Mention, USU Concerto Competition.
2017–2021	USU Presidential Scholarship.
2017–2019	USU Department of Music Scholarship.

Teaching, Mentoring, and Service

Teaching Assistant

- Purdue University — MA 41690 Data Science Lab (Probability).
- Purdue University — Linear Algebra Lab (TA + course development), Spring/Fall 2024.
- Utah State University — PHYS 4650 Optics I, Fall 2020.
- Utah State University — PHYS 2110 Introductory Physics.

Mentoring

Summer 2025 — Undergraduate Research Experience.
Worked with a student to build: cosublimation chamber; inexpensive photon counter.

Summer 2024 — Sloan PReMMiSS Program — Research mentor.

Fall 2019 — Native American STEM Mentorship Program — Mentor (solid-state physics & micro-fabrication).

2014 — ESL Tutor (Spring/Fall 2014).

Other Service

Jan 2026 — Session Chair, “Sources II,” SPIE Photonics West.